

Table formats and the lakehouse

The topic:

- ▶ With no table format in place, the only way to know what's in the folder is to list it and read every file
- ▶ A writer adding new files during that same listing can leave the reader with a partial result -> some new files included, some missing, blending two different moments in time
- ▶ Iceberg keeps a metadata file listing the exact files in each snapshot -> reader stays on one consistent snapshot while the writer builds the next one
- ▶ This same metadata layer lets the schema change (ex.: add a column) without breaking a query written before the change

The questions:

- ▶ How do you turn a folder of files into something that reads like a reliable table, even while it is being written to?
- ▶ How does that same metadata layer also let the table's schema evolve without breaking queries already written?

The demo:

- ▶ Append to a plain Parquet folder while a reader is running and show it read an inconsistent mix. Do the same through an Iceberg table and show the reader get one consistent snapshot. Finish by adding a column and re-running an old query to show it still works.